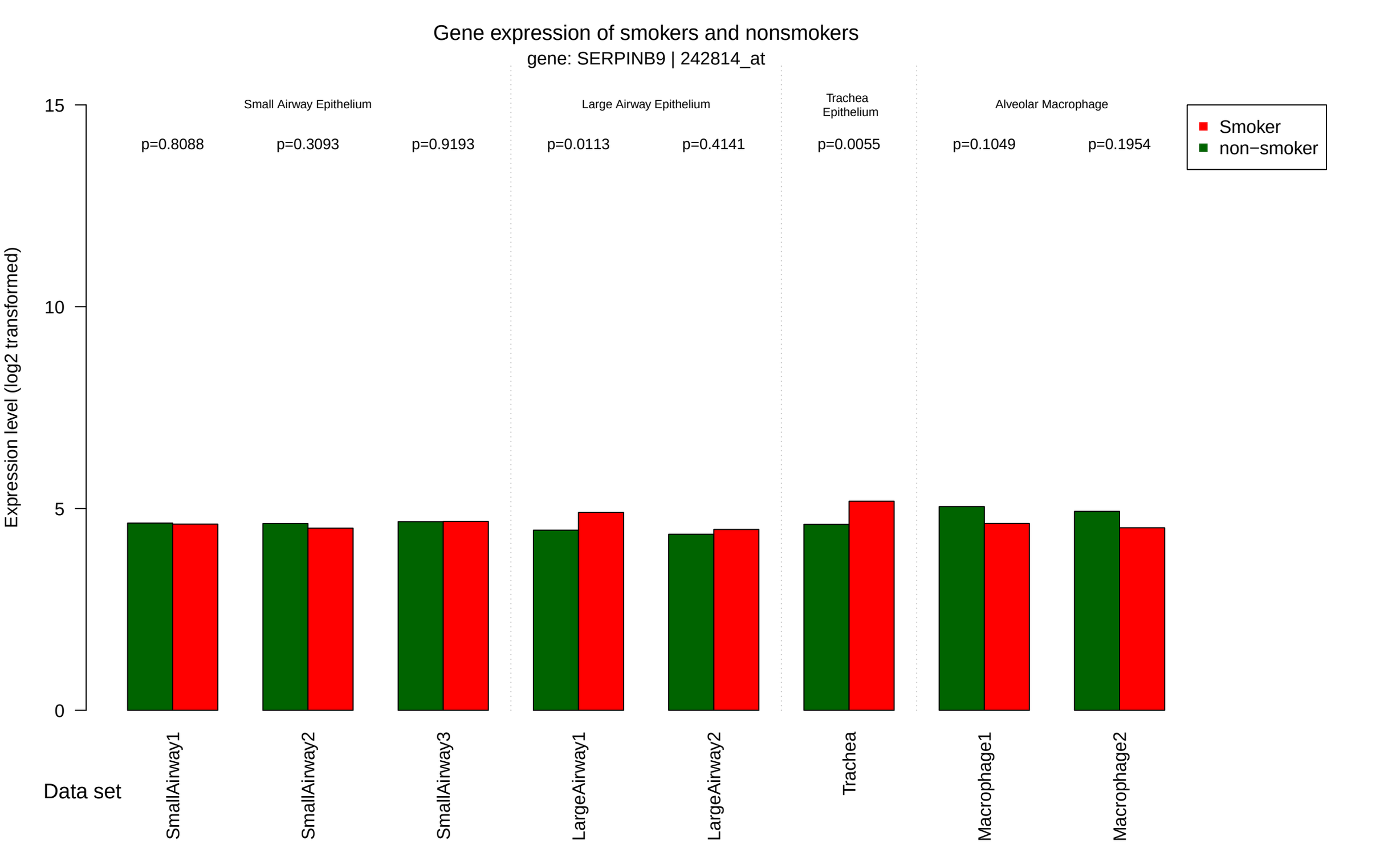
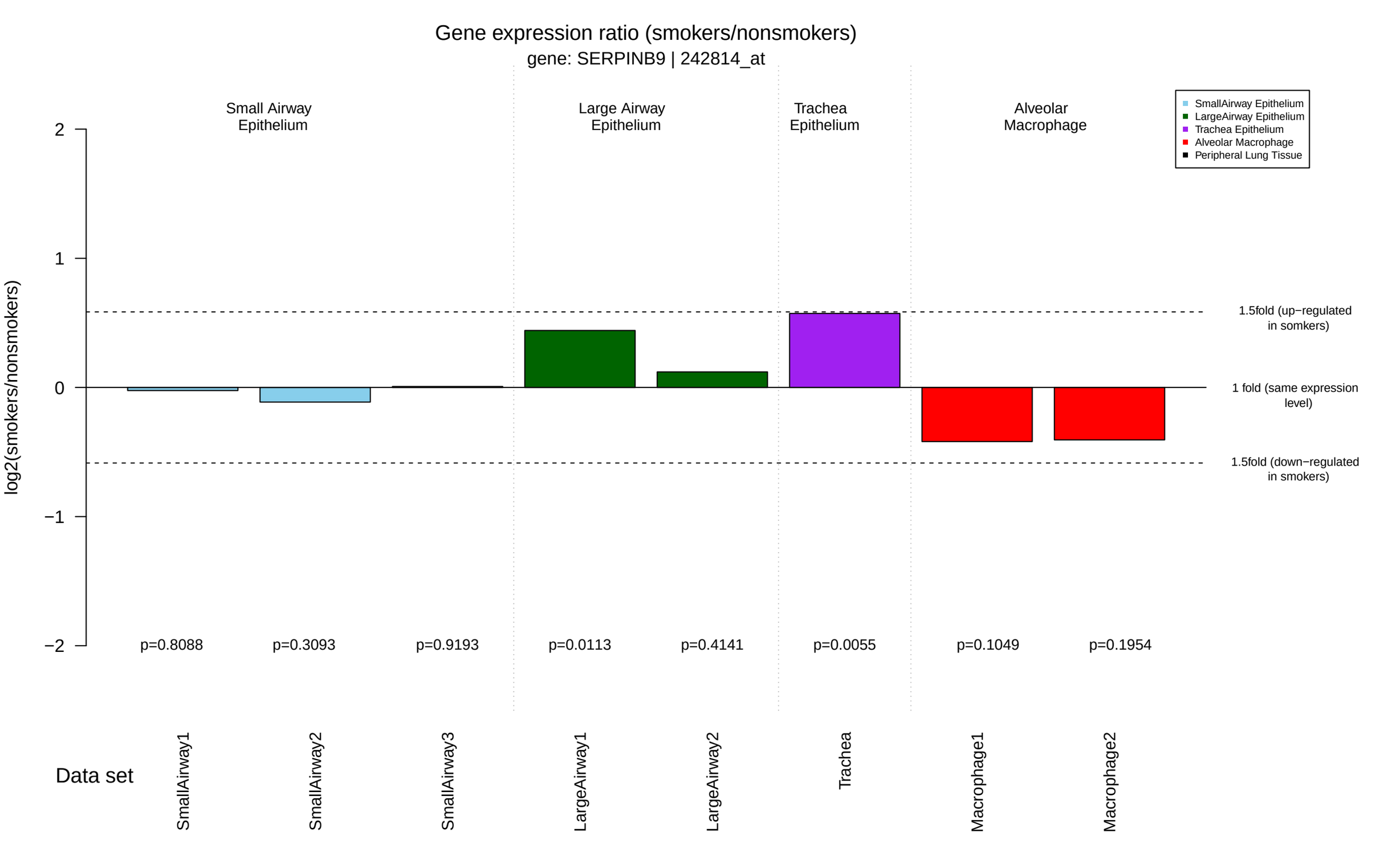
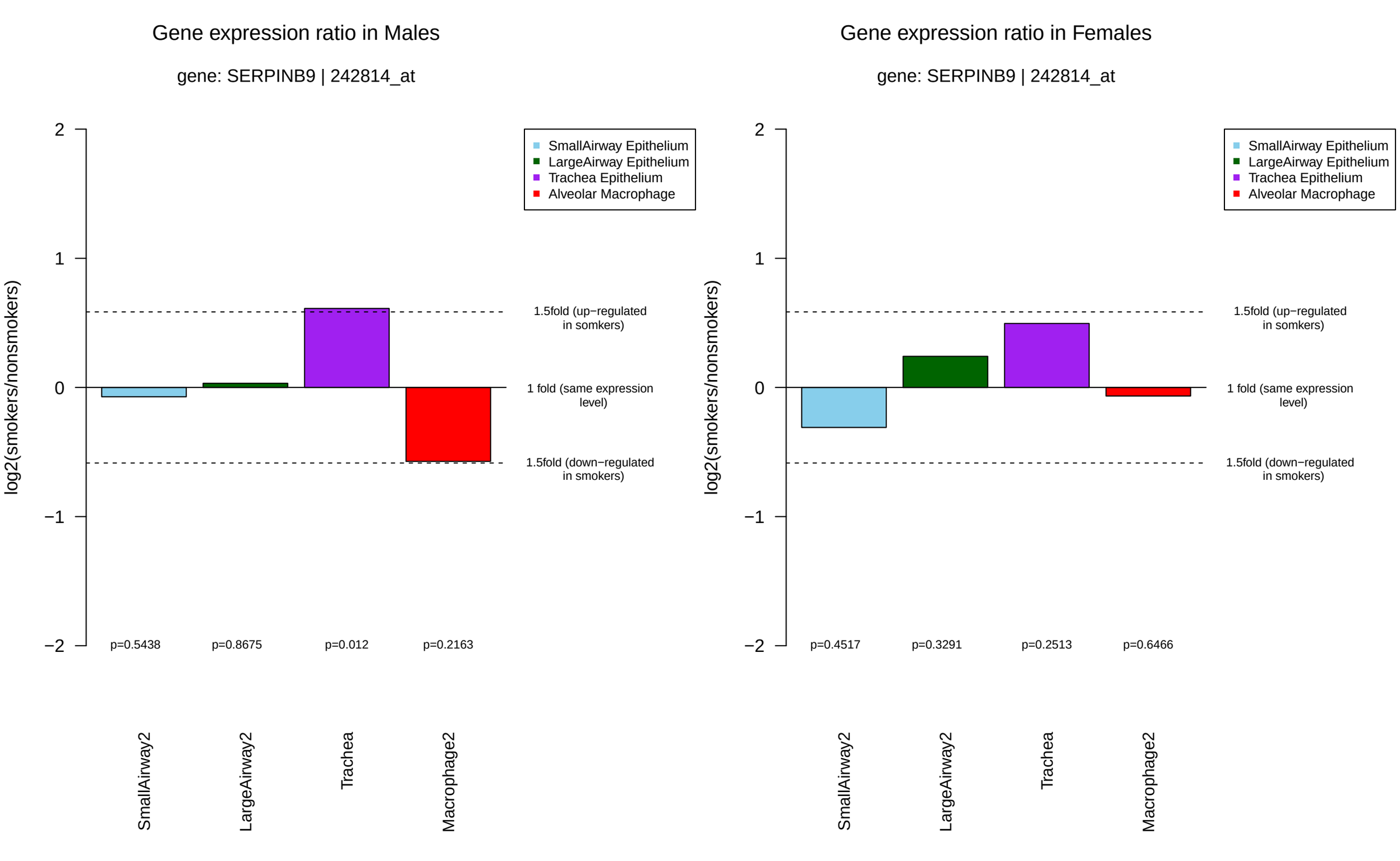
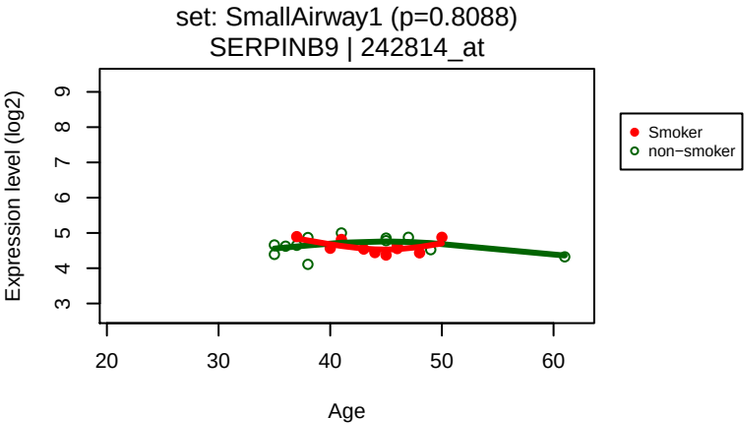
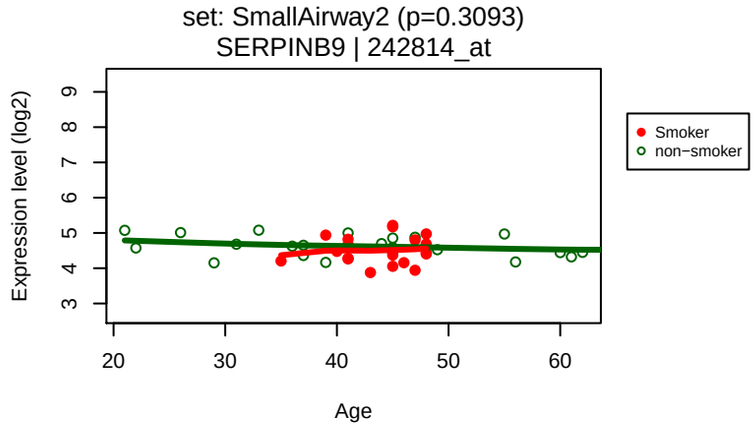
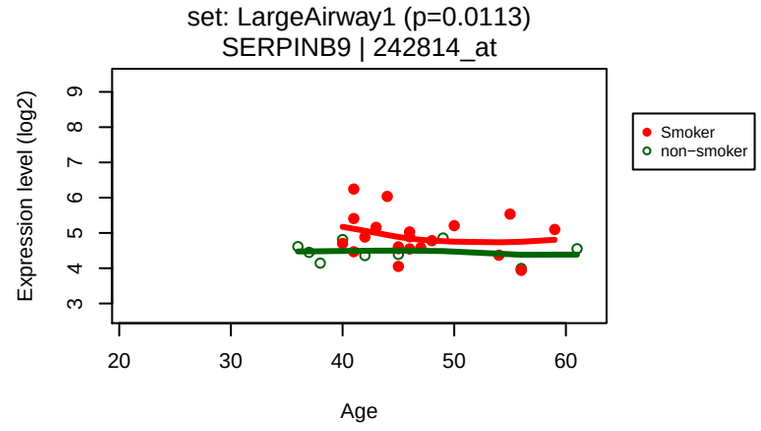

Gene expression of smokers and nonsmokers
gene: SERPINB9 | 242814_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8088
p=0.3093
p=0.9193
p=0.0113
p=0.4141
p=0.0055
p=0.1049
p=0.1954
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

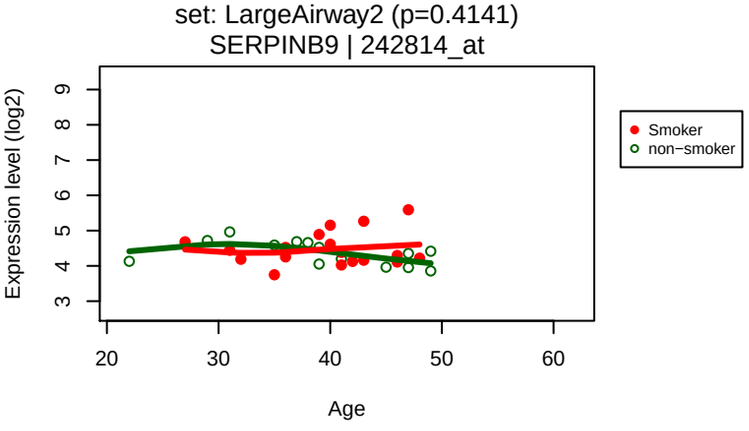

Gene expression ratio (smokers/nonsmokers)
gene: SERPINB9 | 242814_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8088
p=0.3093
p=0.9193
p=0.0113
p=0.4141
p=0.0055
p=0.1049
p=0.1954
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

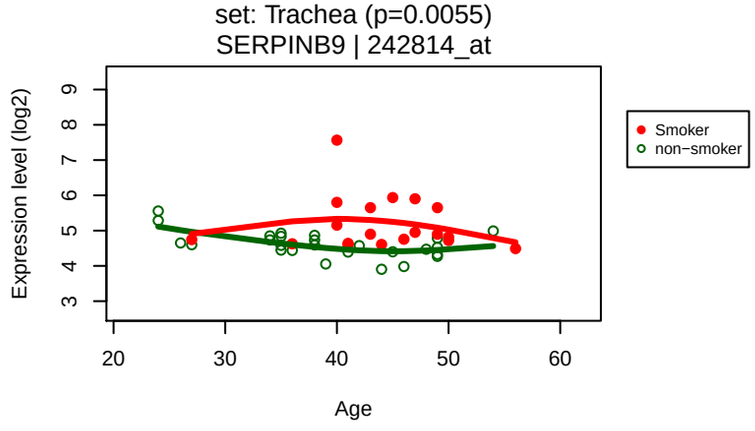

Gene expression ratio in Males
gene: SERPINB9 | 242814_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5438
p=0.8675
p=0.012
p=0.2163
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SERPINB9 | 242814_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4517
p=0.3291
p=0.2513
p=0.6466
SmallAirway2
LargeAirway2
Trachea
Macrophage2

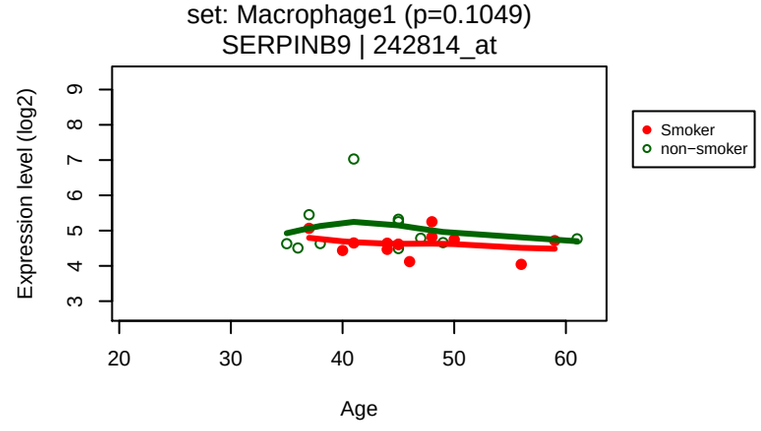

set: SmallAirway1 (p=0.8088)
SERPINB9 | 242814_at
Expression level (log2)
Age
Smoker
non−smoker

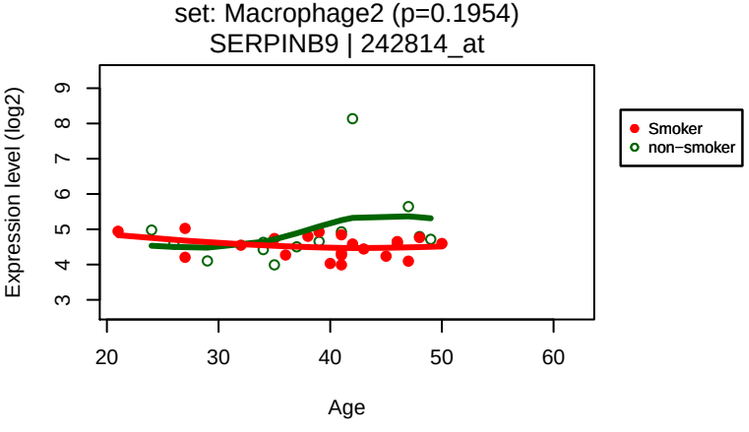

set: SmallAirway2 (p=0.3093)
SERPINB9 | 242814_at
Expression level (log2)
Age
Smoker
non−smoker

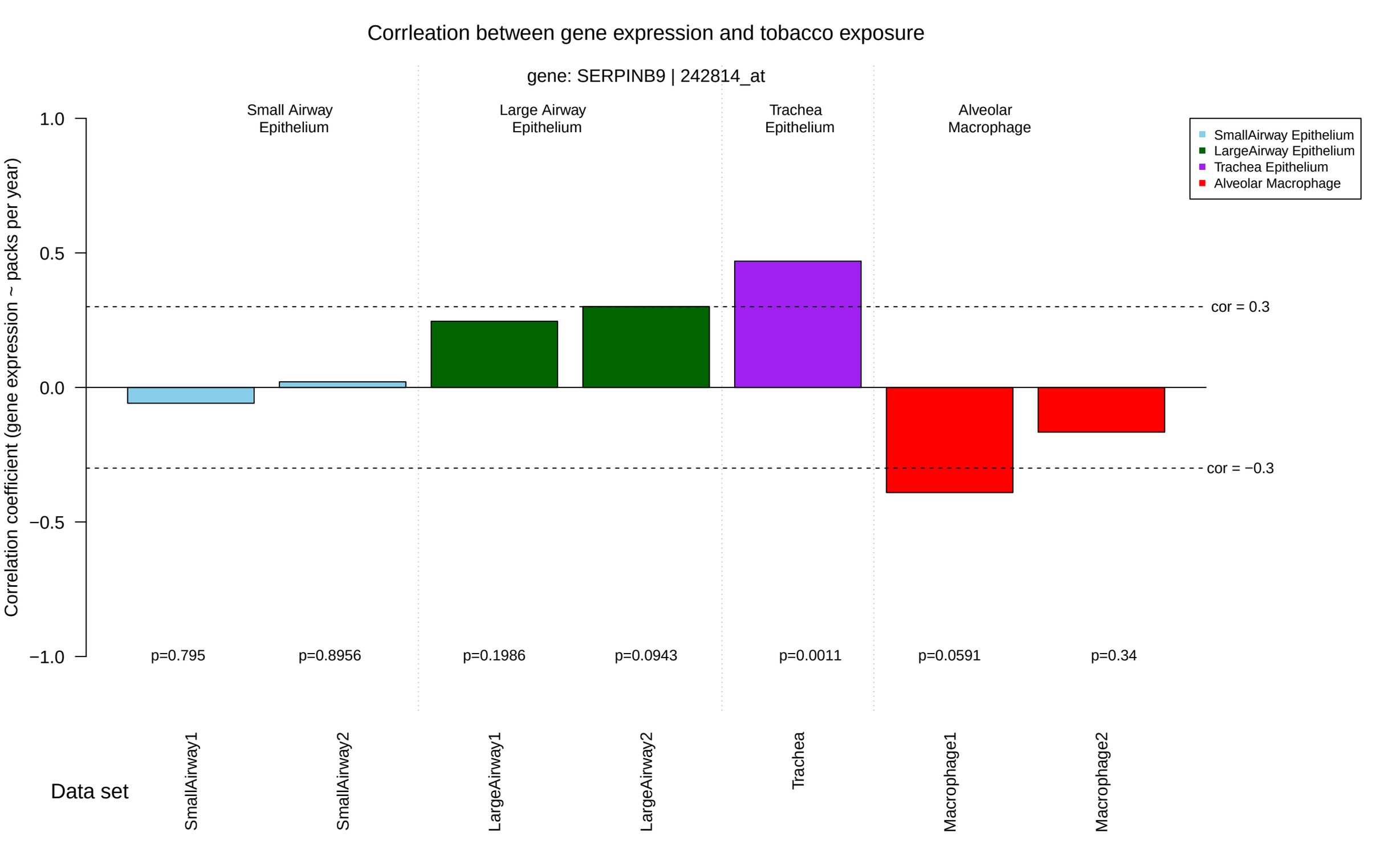

set: LargeAirway1 (p=0.0113)
SERPINB9 | 242814_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4141)
SERPINB9 | 242814_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0055)
SERPINB9 | 242814_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1049)
SERPINB9 | 242814_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1954)
SERPINB9 | 242814_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SERPINB9 | 242814_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.795
p=0.8956
p=0.1986
p=0.0943
p=0.0011
p=0.0591
p=0.34
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2